

Md.Shalha Mucha Bhuyan

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Profile Summary

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MLOps & DevOps Engineer with hands-on experience building and operating scalable, production-grade AI, and full-stack systems across diverse technology stacks. Experienced in designing automated CI/CD pipelines, managing containerized workloads, and optimizing cloud infrastructure for high availability and performance. Focused on optimizing AI lifecycle processes, microservice architecture, reducing deployment time, and ensuring efficient, reliable delivery to end users.

Experience

Machine Learning Ops Engineer

November, 2025 – Present

AKIJ Resource, Dhaka 🔗

- Designed and delivered an end-to-end agentic AI platform using LangGraph for analyzing international market trends and generating real-time business reports, built and orchestrated multi-agent LLM workflows supporting 700+ users across multiple SBUs, reducing decision-making time by 60%.
- Designed and implemented production-grade RESTful APIs using FastAPI, enabling seamless integration and efficient data exchange across microservices.
- Designed and implemented CI/CD pipelines using Jenkins and GitHub Actions, reducing deployment time by 30% and improving release reliability.
- Architected and deployed production-grade cloud infrastructure on AWS (ECR, EKS), containerizing application workloads with Docker for secure image management, automated deployment, and fault-tolerant scalable service delivery across environments.
- Automated server provisioning and configuration management using Ansible, streamlining environment setup, and enforcing consistency across infrastructure.
- Developed Infrastructure as Code using Terraform and CloudFormation, enabling consistent environment provisioning and reducing setup time by 70%.
- Automated AWS operations and cost optimization using Bash and serverless solutions, cutting cloud costs by 20–25%.

Associate Software Engineer (AI/ML)

April – October, 2025

Cloudly Infotech Limited, Dhaka 🔗

- Developed **Maveric Copilot**, a telecommunications-focused agentic orchestration system for real-time network log analysis and automated report generation.
- Developed agentic AI workflows and orchestration pipelines using frameworks such as LangChain, CrewAI, and LLM-based task automation, enabling dynamic multi-agent collaboration and seamless tool integration.
- Fine-tuned LLMs for domain-specific business tasks to improve task performance and relevance.

Education

Chittagong University of Engineering and Technology (CUET), Bangladesh

2019 – 2024

B.Sc in Electronics and Telecommunication Engineering (ETE)

- CGPA 3.20/4.0

Technical Skills

Programming Languages:	Python, C, C++, JavaScript, SQL
Database Management:	PostgreSQL, Vector Databases, Graph Database(Neo4j),MongoDB
Machine Learning & AI:	TensorFlow, PyTorch, LangChain, CrewAI
Backend :	Django, jwt, OAuth
DevOps:	Docker, Jenkins, K8s, Terraform, Bash Scripting, Ansible,Prometheus
MLOps:	MLflow, DVC, DagsHub, Hugging face
Cloud & Infrastructure:	AWS (IAM, S3, EC2, ECR, EKS, ECS, Lambda, VPC, CloudFront)
Linux:	Process Management, Networking, Permissions, File systems, Cron Jobs, Package Management,Log Management,SSH,Systemd
Tools & Platforms:	Git, GitHub, GitLab

Personal Projects

Customer Sentiment Classification

[\[GitHub\]](#) 

- Developed a production-grade full-stack MLOps prediction system to predict customer behavior using multiple machine learning algorithms.
- Deployed this system as a Dockerized using Flask with CI/CD via GitHub Actions on AWS ECR and EKS. Implemented monitoring with Prometheus and Grafana to ensure high availability, low latency, and proactive issue detection.
- **Tools Used:** AWS (EKS, ECR, S3, EC2), MLflow, DVC, Docker, Flask, Scikit-Learn.

Cloud-Native Application Deployment on AWS

- Deployed a containerized web application on AWS EKS using serverless compute, eliminating node management and improving operational efficiency.
- Configured Application Load Balancer (ALB) integration for external access, enabling secure and reliable traffic routing.

Certifications

AWS Cloud Technical Essentials (AWS), **Agentic AI** (DeepLearning.ai), **Fundamentals of MCP** (Hugging Face), **Generative AI with Large Language Models** (Coursera), **TensorFlow Developer Professional Certificate** (Coursera)

Soft Skills

- Problem-Solving, Communication, Team Collaboration, Mentoring